Replay-Aware TSP Benchmark Report

Overview

- Condition count: 9.
- Best final transfer gap: phase6_no_replay.
- Best TSPLIB holdout gap: phase6_no_replay.
- Best synthetic holdout gap: phase6_random_replay.

Run Metadata

- run_name: run_20260519_033027_p
- started_at_local: 2026-05-19 03:30:27
- finished_at_local: 2026-05-19 03:49:26
- duration_hhmm: 00:19
- duration_seconds: 1138.716
- seed_offset: 15000
- replicate_label: p
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final TSPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Archive Diversity	Archive Hardness	Failure Concentration	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
phase6_no_replay	none	score_only	False	0.070305	0.015447	0.050357	0.236772	0.185541	0.0	0.651562	0.76	0.063188
phase6_random_replay	random	score_only	False	0.134742	0.0	0.085745	0.236772	0.141857	0.19457	0.0	0.76	0.0
phase6_failure_replay	failure	score_only	False	0.11596	0.0	0.073793	0.171206	0.32623	0.214526	0.77161	0.76	0.037481

Condition	Replay	Selection	Compression	Final TSPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Archive Diversity	Archive Hardness	Failure Concentration	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
phase6_random_replay_compression	random	novelty_gate	True	0.213387	0.064916	0.159398	0.236772	0.214847	0.19457	0.0	0.56	0.0
phase6_stratified_random_replay	stratified_random	score_only	False	0.153484	0.0	0.097672	0.236772	0.161596	0.221162	0.0	0.76	0.0
phase6_diversity_weighted_replay	diversity_weighted	score_only	False	0.158723	0.0	0.101006	0.236772	0.165022	0.157119	0.0	0.76	0.0
phase6_residual_failure_replay	residual_failure	score_only	False	0.213387	0.064916	0.159398	0.149274	0.338865	0.291666	0.850599	0.56	-0.029279
phase6_diversity_failure_replay	diversity_failure	score_only	False	0.213387	0.064916	0.159398	0.171206	0.348017	0.17024	0.856496	0.56	-0.029078
phase6_diversity_failure_replay_compression	diversity_failure	novelty_gate	True	0.213387	0.064916	0.159398	0.171206	0.348017	0.17024	0.0	0.56	0.0

Condition Notes

phase6_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: TSPLIB 0.070305, synthetic 0.015447, combined 0.050357.
- Accepted-epoch count 3, mean accepted code novelty 0.651562, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.185541, size bias 8.3e-05, and failure concentration 0.0.
- Replay selection diversity 0.0, mean selected expected gap 0.0, mean selected residual gap 0.0.
- Adaptation efficiency 0.063188 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 5, 'adversarial_layouts': 4} / elite 3.
- Panel heldout_tsplib mean gap 0.070305 across 7 instances; family means: ch=0.060074, kroD=0.085282, pcb=0.236283, pr=0.004882, rd=0.020354, st=0.025185.
- Panel synthetic_holdout mean gap 0.015447 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.061786, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3409, "expected_gap": 0.40822, "family": "a", "name": "a280", "optimality_gap": 0.32183, "residual_gap": -0.08639}, {"best_known_cost": 629, "cost": 827, "expected_gap": 0.257552, "family": "eil", "name": "eil101", "optimality_gap": 0.314785, "residual_gap": 0.057233},

{"best_known_cost": 14379, "cost": 16308, "expected_gap": 0.32496, "family": "lin", "name": "lin105", "optimality_gap": 0.134154, "residual_gap": -0.190806}]

phase6_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: TSPLIB 0.134742, synthetic 0.0, combined 0.085745.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.141857, size bias 8.3e-05, and failure concentration 0.19457.
- Replay selection diversity 0.262174, mean selected expected gap 0.133408, mean selected residual gap -0.009805.
- Adaptation efficiency 0.0 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.134742 across 7 instances; family means: ch=0.120922, kroD=0.122805, pcb=0.234629, pr=0.216154, rd=0.072946, st=0.054815.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 14379, "cost": 18560, "expected_gap": 0.321761, "family": "lin", "name": "lin105", "optimality_gap": 0.290771, "residual_gap": -0.03099}, {"best_known_cost": 2579, "cost": 3183, "expected_gap": 0.403955, "family": "a", "name": "a280", "optimality_gap": 0.234199, "residual_gap": -0.169756}, {"best_known_cost": 7542, "cost": 8126, "expected_gap": 0.236622, "family": "berlin", "name": "berlin52", "optimality_gap": 0.077433, "residual_gap": -0.159189}]

phase6_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.11596, synthetic 0.0, combined 0.073793.
- Accepted-epoch count 4, mean accepted code novelty 0.77161, and final complexity 0.76.
- Active replay archive worst_archive with mean diversity 0.171206, mean hardness 0.32623, size bias 0.120322, and failure concentration 0.214526.
- Replay selection diversity 0.204223, mean selected expected gap 0.316883, mean selected residual gap 0.032031.
- Adaptation efficiency 0.037481 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 5, 'adversarial_layouts': 4} / elite 4.
- Panel heldout_tsplib mean gap 0.11596 across 7 instances; family means: ch=0.102726, kroD=0.084296, pcb=0.272736, pr=0.119186, rd=0.081163, st=0.048889.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3269, "expected_gap": 0.403722, "family": "a", "name": "a280", "optimality_gap": 0.267546, "residual_gap": -0.136176}, {"best_known_cost": 629, "cost": 676, "expected_gap": 0.25469, "family": "eil", "name": "eil101", "optimality_gap": 0.074722, "residual_gap": -0.179968}, {"best_known_cost": 7542, "cost": 7990, "expected_gap": 0.233148, "family": "berlin", "name": "berlin52", "optimality_gap": 0.059401, "residual_gap": -0.173747}]

phase6_random_replay_compression

- Replay mode: random with selection mode novelty_gate.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.

- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.56.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.214847, size bias 8.3e-05, and failure concentration 0.19457.
- Replay selection diversity 0.262174, mean selected expected gap 0.132217, mean selected residual gap 0.039149.
- Adaptation efficiency 0.0 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "expected_gap": 0.402947, "family": "a", "name": "a280", "optimality_gap": 0.401318, "residual_gap": -0.001629}, {"best_known_cost": 629, "cost": 875, "expected_gap": 0.253418, "family": "eil", "name": "eil101", "optimality_gap": 0.391097, "residual_gap": 0.137679}, {"best_known_cost": 14379, "cost": 19107, "expected_gap": 0.323194, "family": "lin", "name": "lin105", "optimality_gap": 0.328813, "residual_gap": 0.005619}]

phase6_stratified_random_replay

- Replay mode: stratified_random with selection mode score_only.
- Final transfer gaps: TSPLIB 0.153484, synthetic 0.0, combined 0.097672.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.161596, size bias 8.3e-05, and failure concentration 0.221162.
- Replay selection diversity 0.187367, mean selected expected gap 0.236357, mean selected residual gap -0.020462.
- Adaptation efficiency 0.0 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.153484 across 7 instances; family means: ch=0.042933, kroD=0.239034, pcb=0.287565, pr=0.295648, rd=0.141087, st=0.025185.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3389, "expected_gap": 0.402171, "family": "a", "name": "a280", "optimality_gap": 0.314075, "residual_gap": -0.088096}, {"best_known_cost": 14379, "cost": 18147, "expected_gap": 0.322373, "family": "lin", "name": "lin105", "optimality_gap": 0.262049, "residual_gap": -0.060324}, {"best_known_cost": 629, "cost": 754, "expected_gap": 0.25628, "family": "eil", "name": "eil101", "optimality_gap": 0.198728, "residual_gap": -0.057552}]

phase6_diversity_weighted_replay

- Replay mode: diversity_weighted with selection mode score_only.
- Final transfer gaps: TSPLIB 0.158723, synthetic 0.0, combined 0.101006.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.165022, size bias 8.3e-05, and failure concentration 0.157119.
- Replay selection diversity 0.269935, mean selected expected gap 0.1547, mean selected residual gap 0.001098.

- Adaptation efficiency 0.0 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 4, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.158723 across 7 instances; family means: ch=0.091954, kroD=0.295858, pcb=0.209835, pr=0.240627, rd=0.151201, st=0.02963.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 14379, "cost": 17755, "expected_gap": 0.321803, "family": "lin", "name": "lin105", "optimality_gap": 0.234787, "residual_gap": -0.087016}, {"best_known_cost": 2579, "cost": 3183, "expected_gap": 0.402947, "family": "a", "name": "a280", "optimality_gap": 0.234199, "residual_gap": -0.168748}, {"best_known_cost": 7542, "cost": 8264, "expected_gap": 0.233068, "family": "berlin", "name": "berlin52", "optimality_gap": 0.095731, "residual_gap": -0.137337}]

phase6_residual_failure_replay

- Replay mode: residual_failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 3, mean accepted code novelty 0.850599, and final complexity 0.56.
- Active replay archive residual_archive with mean diversity 0.149274, mean hardness 0.338865, size bias 0.093558, and failure concentration 0.291666.
- Replay selection diversity 0.177489, mean selected expected gap 0.264336, mean selected residual gap 0.088057.
- Adaptation efficiency -0.029279 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "expected_gap": 0.405273, "family": "a", "name": "a280", "optimality_gap": 0.401318, "residual_gap": -0.003955}, {"best_known_cost": 629, "cost": 875, "expected_gap": 0.262003, "family": "eil", "name": "eil101", "optimality_gap": 0.391097, "residual_gap": 0.129094}, {"best_known_cost": 14379, "cost": 19107, "expected_gap": 0.321511, "family": "lin", "name": "lin105", "optimality_gap": 0.328813, "residual_gap": 0.007302}]

phase6_diversity_failure_replay

- Replay mode: diversity_failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 3, mean accepted code novelty 0.856496, and final complexity 0.56.
- Active replay archive worst_archive with mean diversity 0.171206, mean hardness 0.348017, size bias 0.120322, and failure concentration 0.17024.
- Replay selection diversity 0.199284, mean selected expected gap 0.301618, mean selected residual gap 0.048371.
- Adaptation efficiency -0.029078 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.

- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "expected_gap": 0.405351, "family": "a", "name": "a280", "optimality_gap": 0.401318, "residual_gap": -0.004033}, {"best_known_cost": 629, "cost": 875, "expected_gap": 0.267091, "family": "eil", "name": "eil101", "optimality_gap": 0.391097, "residual_gap": 0.124006}, {"best_known_cost": 14379, "cost": 19107, "expected_gap": 0.321511, "family": "lin", "name": "lin105", "optimality_gap": 0.328813, "residual_gap": 0.007302}]

phase6_diversity_failure_replay_compression

- Replay mode: diversity_failure with selection mode novelty_gate.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.56.
- Active replay archive worst_archive with mean diversity 0.171206, mean hardness 0.348017, size bias 0.120322, and failure concentration 0.17024.
- Replay selection diversity 0.199284, mean selected expected gap 0.300417, mean selected residual gap 0.049571.
- Adaptation efficiency 0.0 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "expected_gap": 0.405739, "family": "a", "name": "a280", "optimality_gap": 0.401318, "residual_gap": -0.004421}, {"best_known_cost": 629, "cost": 875, "expected_gap": 0.263911, "family": "eil", "name": "eil101", "optimality_gap": 0.391097, "residual_gap": 0.127186}, {"best_known_cost": 14379, "cost": 19107, "expected_gap": 0.321274, "family": "lin", "name": "lin105", "optimality_gap": 0.328813, "residual_gap": 0.007539}]

Judge Appendix

Analysis of Replay-Aware TSP Benchmark Suite Results

Summary

- **Best holdout TSPLIB gap:** phase6_no_replay (0.0703)
- **Best synthetic holdout gap:** phase6_random_replay (0.0)
- **Best overall transfer condition:** phase6_no_replay (transfer gap 0.0504)
- **Total conditions analyzed:** 9

Key Metrics per Condition (mean gaps lower is better)

Condition	Synthetic Holdout Gap	TSPLIB Holdout Gap	Transfer Gap	Replay Mode	Code Novelty (mean)
phase6_no_replay	0.0154	**0.0703 (best)**	0.0504 (best)	none	0.6516
phase6_random_replay	**0.0 (best)**	0.1347	0.0857	random	0.0

Condition	Synthetic Holdout Gap	TSPLIB Holdout Gap	Transfer Gap	Replay Mode	Code Novelty (mean)
phase6_failure_replay	**0.0**	0.1160	0.0738	failure	0.7716
phase6_random_replay_compression	0.0649	0.2134	0.1594	random	0.0
phase6_stratified_random_replay	0.0	0.1535	0.0977	stratified_random	0.0
phase6_diversity_weighted_replay	0.0	0.1587	0.1010	diversity_weighted	0.0
phase6_residual_failure_replay	0.0649	0.2134	0.1594	residual_failure	0.8506
phase6_diversity_failure_replay	0.0649	0.2134	0.1594	diversity_failure	0.8565
phase6_diversity_failure_replay_compression	0.0649	0.2134	0.1594	diversity_failure	0.0

Interpretation of Main Transfer and Optimality-Gap Evidence

- **Transfer performance (TSPLIB gap and synthetic holdout gap):**
- The no-replay condition phase6_no_replay achieves the best transfer gap (0.0504) and lowest TSPLIB holdout gap (0.0703), suggesting strongest generalization.
- Random replay (phase6_random_replay) achieves perfect synthetic holdout gap (0.0) but has significantly worse TSPLIB gap (0.1347) and transfer gap (0.0857).
- All replay-based conditions with compression or diversity weighting show *worse* transfer gaps (~0.1-0.16) and TSPLIB gaps (≥0.15), indicating reduced transfer despite varied replay strategies.
- Failure-focused replay modes (phase6_failure_replay) have intermediate performance (transfer gap 0.0738, TSPLIB gap 0.1160), better than diversity/compression replay.

- **Code novelty:**
- Conditions with replay have near-zero code novelty, except phase6_no_replay (mean 0.6516) and failure replay modes (phase6_failure_replay and phase6_residual_failure_replay, ~0.77-0.85).
- Compression and diversity-based conditions have zero code novelty despite worse transfer, indicating no new code inventions under those replay modes.

Archive and Replay Mechanism Insights

Condition	Archive Key	Archive Descriptor Diversity	Archive Hardness	Archive Size Bias	Replay Failure Concentration	Replay Selection Diversity
phase6_no_replay	experience_archive	0.240	0.204	0.0	0.0	0.0
phase6_random_replay	experience_archive	0.237	0.142	8.3e-05	0.195 (final)	0.262
phase6_failure_replay	worst_archive	0.172	0.335	0.125	0.243	0.204

Condition	Archive Key	Archive Descriptor Diversity	Archive Hardness	Archive Size Bias	Replay Failure Concentration	Replay Selection Diversity
phase6_random_replay_compression	experience_archive	0.237	0.212	0.0	0.195	0.262
phase6_stratified_random_replay	experience_archive	0.237	0.188	0.0	0.170	0.269
phase6_diversity_weighted_replay	experience_archive	0.240	0.197	0.0	0.105	0.270
phase6_residual_failure_replay	residual_archive	0.150	0.339	0.094	0.292 (mean)	0.177
phase6_diversity_failure_replay	worst_archive	0.175	0.349	0.125	0.170	0.199
phase6_diversity_failure_replay_compression	worst_archive	0.175	0.349	0.125	0.170	0.199

- Replay failure concentration is zero for no replay (as expected).
- Failure replay and residual failure replay have highest failure concentration (≥0.24), focusing on harder cases.
- Diversity-weighted and stratified replay have moderate replay selection diversity (~0.2-0.27), indicating efforts to diversify, but still don't improve transfer.
- Archive descriptor diversity is highest (~0.24) in no replay and experience archives; worst and residual archives have lower diversity (~0.15-0.17).
- Archive hardness is notably higher in failure and residual replay modes (>0.33) compared to no-replay (~0.20) or random playback (~0.14).

Conservative Conclusions

- **Transfer performance is strongest without replay (phase6_no_replay), with lowest TSPLIB and transfer gaps, despite moderate synthetic holdout gap.**
- Replay strategies, including random, stratified, diversity-weighted, and failure-based replay, do not improve transfer performance compared to no replay.
- Synthetic holdout gap is minimized (even zero) for random and diversity replay modes but this does not translate into improved transfer or TSPLIB generalization.
- Code novelty is *not* correlated with better transfer; notably, zero code novelty replay conditions (e.g., random replay) have worse transfer.
- Failures concentrate more with failure-based replay modes, increasing archive hardness but without corresponding transfer improvements.
- Compression-aware replay reduces complexity but also reduces transfer and increases gaps.
- Overall, replay variants do not show consistent benefit over no replay; replay introduces data biases (e.g., size bias, failure concentration) not beneficial for transfer.
- Lexical code novelty reduction under replay does not imply algorithmic innovation; measured adaptations appear limited or negative on transfer metrics.

Summary:

- **Top transfer and TSPLIB holdout performance observed for phase6_no_replay (no replay), with low transfer gaps (~0.05-0.07).**
- **Random replay and other replay modes achieve better synthetic holdout performance but at cost of worse transfer and TSPLIB gaps.**
- **Failure replay modes yield more focused/harder archives but don't improve transfer compared to no replay.**
- **Replay reduces code novelty but no algorithmic advantage is evident as per gap metrics.**
- **Diversified or compression-aware replay does not improve generalization or replay benefit per transfer gaps and archive metrics.**

Hence, monitoring optimality gap and transfer gaps suggests no replay is preferable for transfer, while replay strategies often hurt or fail to enhance transfer despite archival or code impact.